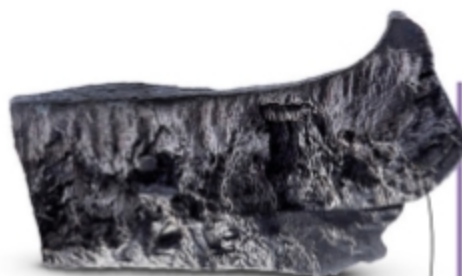


Gadolinium

64 Gd Named after the Finnish chemist Johan Gadolin, gadolinium can be used to treat tumors in a treatment called neutron therapy. Alloys made with small amounts of this element can resist high temperatures.



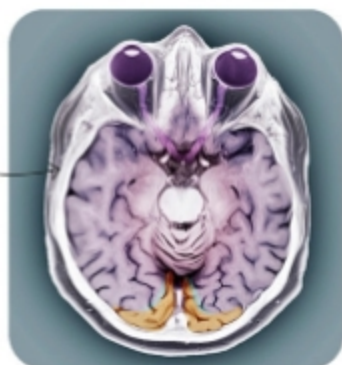
Laboratory sample of pure gadolinium *Soft, silvery metal*



Gadolinium MRI scan of a healthy human brain and eyes

Gadolinite

This black mineral only contains very tiny amounts of gadolinium. Its composition varies but usually has cerium or yttrium mixed with other elements.



Clear images

A non-toxic form of gadolinium is used in MRI (magnetic resonance imaging) scans as a contrasting agent to help improve the quality of the images of internal body structures.

Terbium

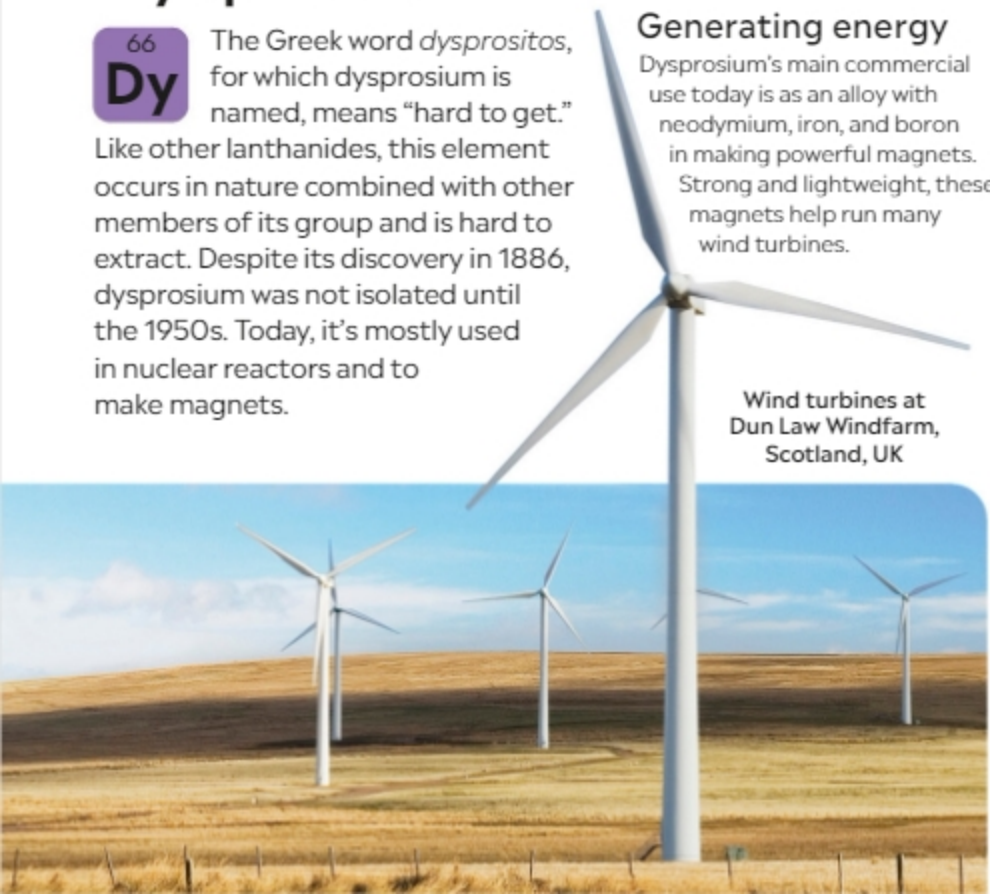
65 Tb Terbium was discovered by Swedish chemist Carl Mosander, who detected it as an impurity in yttrium oxide. It's one of four elements named after the tiny Swedish village of Ytterby—the others are erbium, yttrium, and ytterbium, and all were isolated from gadolinite, which was mined there. Because of their similarities, the elements were confused in the 19th century, and the names of erbium and terbium ended up switched. Today, terbium is used in fuel cells, in special alloys, and to make some types of fluorescent lamps.

Dysprosium

66 Dy The Greek word *dysprositos*, for which dysprosium is named, means "hard to get." Like other lanthanides, this element occurs in nature combined with other members of its group and is hard to extract. Despite its discovery in 1886, dysprosium was not isolated until the 1950s. Today, it's mostly used in nuclear reactors and to make magnets.

Generating energy

Dysprosium's main commercial use today is as an alloy with neodymium, iron, and boron in making powerful magnets. Strong and lightweight, these magnets help run many wind turbines.

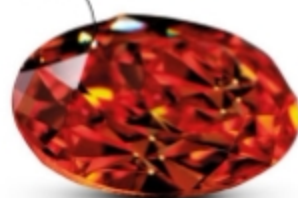


Wind turbines at Dun Law Windfarm, Scotland, UK

Holmium

67 Ho This element is a relatively unreactive, malleable, silvery-white metal. Holmium has the highest magnetic strength of any metal, so it's used to make all kinds of magnets.

The red color in this zirconia gemstone comes from holmium oxide.



Rich color

Holmium impurities can give a red or yellow color to a zirconia gemstone. The compound holmium oxide can also be used to color glass.